

Dohyun (Eric) Kim

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EDUCATION

University of Auckland - Auckland CBD, Auckland | Expected Graduation **Nov 2027**

- **Degree:** Bachelor's (Hons) of Engineering in Computer Systems Engineering
- **Concentration:** Embedded systems & CSE with strong software integration
- **Related Coursework:** Computer Systems Design, Object Oriented Programming, Electronics, Electrical Systems, Mathematical Modelling, Software Quality Assurance

PROJECTS

"Winnie the Bot" (3rd Place) – ECSE Design Competition | Team Project | 2025

- Built an **interactive robot** featuring face recognition, speech input and output, servo based movements, and ChatGPT powered dialogue, and contributed to hardware and software integration and exhibition materials
- Delivered a functional human-machine interaction prototype integrating face recognition, speech I/O, servo control, and ChatGPT-powered dialogue, earning 3rd place among competing teams

Web Design – KEB Project Playground | Team Project | 2025

- Developed a **React based website** for the Korean Engineering Body KEB to showcase events and executive members and collaborated with senior students to design UI components using Figma and implement interactive layouts
- Implemented responsive UI components using JavaScript and React, and designed interactive layouts in Figma for a production website now used by KEB members

Smart Energy Monitor – Design Project | Team Project | 2025

- Designed an **embedded system** to measure and display energy usage using ATmega microcontrollers and worked on sensor interfacing ADC data handling and signal conditioning
- Produced a validated PCB schematic using Altium Designer and LTspice simulations, achieving accurate sensor-to-display signal conditioning across the full hardware stack

Java Programming and OOP Projects – Java | Individual Project | 2025

- Developed multiple **Java command line systems** including an operator management tool an adaptive AI Mind Game and a graph based routing application
- Applied **OOP principles** design patterns such as Strategy and Factory and data structures like List Map Set and Queue while building modular test driven programs using Maven and JUnit and improving Java fluency and problem solving for real world scenarios

RoastWorks Analytics – Python | Team Project | 2026

- Built a Python desktop analytics app (**PyQt6, pandas**) that automated a full-day Excel reporting workflow into under 30 seconds.
- Implemented three time-series **forecasting models** with holdout MAE/RMSE validation and configurable horizons.

EXPERIENCE

Creative Imaginary Lab (ciLab) - Robotics Instructor | Apr 2026 - Present

- Design and deliver hands-on robotics curriculum for students aged 8-16, covering embedded programming, sensor integration, and competition strategy, with multiple student teams advancing to regional events

Twelve Restaurant - Front of House (FOH) | Jul 2024 - Jan 2025

- Delivered high-quality front-of-house service in a high-volume restaurant environment, managing guest experience and coordinating efficient table transitions under pressure

SKILLS

Languages: Java, C, Python, JavaScript

Frameworks & Tools: [React.js](#), Git, Maven, JUnit, Figma, Amazon Web Services (AWS)

Hardware & EDA: Atmel AVR, Altium Designer, LTspice, Arduino IDE, Quartus Prime

Other: MATLAB, VHDL, PyQt

LICENSES & CERTIFICATIONS

Amazon Web Services (AWS) Certified Cloud Practitioner | Apr 2026

- Demonstrated proficiency in AWS core services (EC2, S3, IAM, Lambda, VPC) and cloud architecture principles including the Well-Architected Framework, security best practices, and cost optimisation.

ACTIVITIES AND LEADERSHIP

Korean Engineering Body (KEB) | 2024 - Present

- **Academic Team Executive** - Organize and lead tutorials for first year engineering students to strengthen KEB's academic support and visibility on campus
- **Mentor** - Support first-year students with academic guidance and social integration into university life